Machine Learning – Assignment Answers

What is a parameter?

A parameter is a value that defines a population or a machine learning model. In ML, parameters are learned from data during training, such as weights in linear regression.

What is correlation?

Correlation measures the strength and direction of the relationship between two variables. Its value ranges from -1 to +1.

What does negative correlation mean?

Negative correlation means that as one variable increases, the other decreases.

Define Machine Learning and its components

Machine Learning is a field of AI that allows systems to learn from data. Main components include data, features, model, loss function, optimizer, and evaluation metrics.

How does loss value help determine model quality?

Loss value indicates how far predictions are from actual values. Lower loss means better performance.

What are continuous and categorical variables?

Continuous variables take numeric values like height or weight. Categorical variables represent labels or categories like gender or color.

Handling categorical variables

Categorical variables are handled using techniques like Label Encoding, One-Hot Encoding, and Ordinal Encoding.

Training and testing dataset

Training data is used to train the model, while testing data is used to evaluate performance.

What is sklearn.preprocessing?

It is a module in scikit-learn used for scaling, normalization, and encoding of data.

What is a test set?

A test set is used to evaluate the trained model on unseen data.

How do we split data in Python?

We use train_test_split from sklearn.model_selection to split data into training and testing sets.

Approach to a Machine Learning problem

Understand the problem, collect data, perform EDA, preprocess data, split data, train model, evaluate, and improve.

Why perform EDA before modeling?

EDA helps understand data patterns, detect outliers, missing values, and relationships.

Finding correlation in Python

Correlation can be calculated using df.corr() in pandas.

What is causation?

Causation means one variable directly affects another. Correlation does not imply causation.

What is an optimizer?

An optimizer updates model parameters to minimize loss. Examples include Gradient Descent, SGD, Adam.

What is sklearn.linear_model?

It is a module that provides linear models like LinearRegression and LogisticRegression.

What does model.fit() do?

model.fit() trains the model using features and target values.

What does model.predict() do?

model.predict() generates predictions for new input data.

What is feature scaling?

Feature scaling brings all features to a similar scale to improve model performance.

How do we perform scaling in Python?

Scaling is performed using tools like StandardScaler or MinMaxScaler.

Explain data encoding

Data encoding converts categorical data into numerical format so ML models can process it.